

Descriptor Invariance versus Regressor Choice for a Toy Machine-Learned Potential on Small Lennard-Jones Clusters

Vega-95P

STEM seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/lennard-jones-invariant-descriptors>

Abstract—We study a toy machine-learned potential for 5–13 atom Lennard-Jones clusters (perturbed minima and random configurations) with three input representations (raw coordinates, sorted inverse pair distances, sum-pooled Behler–Parrinello symmetry functions) and two regressors (kernel ridge regression, a small MLP), plus an atomwise Behler–Parrinello-style network, all reimplemented. We measure test energy error, learning curves for 50–1500 training configurations, and error under random rotation and atom permutation of the test set, with 3–5 seeds on CPU. Raw coordinates fail: with randomly rotated and ordered training data, raw KRR is indistinguishable from predicting the mean energy (MAE 12.178 versus 12.203 at $n=500$) and the raw MLP is worse (15.028), so their error does not rise further under rotation or permutation; their predictions change by 2.881 and 15.23 energy units on average under such a transform, while every invariant system stays below 10^{-4} . At $n=500$ the sorted-distance MLP is the most accurate system (0.635), followed by symmetry-function KRR (0.889) and sorted-distance KRR (1.095). Symmetry-function KRR is more accurate than sorted-distance KRR at $n=50$, 150 and 500 (Welch $p = 0.02647$ at $n=500$) and less accurate at $n=1500$ (0.7131 versus 0.4531), which supports our pre-registered small-data hypothesis for KRR, on 3 seeds per size at $n=50$ and 150. This verdict depends on the KRR hyperparameter grid: a narrower grid, used in the first version of this study, clips the kernel width and gives 1.587 at $n=500$. The selected kernels are nearly flat, and plain linear ridge regression on the same descriptor reaches 0.805, whereas the two networks on symmetry functions reach only 4.513 and 3.130. Removing the angular symmetry functions has no resolved effect (0.785 versus 0.889, with overlapping seed ranges). This is a small single-system study with a shared minima library; it supports no claim beyond it.

Index Terms—machine-learned interatomic potentials, symmetry functions, kernel ridge regression, permutation invariance, Lennard-Jones clusters

I. Introduction

Machine-learned interatomic potentials replace expensive quantum-mechanical energy evaluations by a regression model of the potential energy surface [1]. The true energy is invariant under rotation, translation and permutation of identical atoms, so the choice of input

representation decides whether a model must learn these symmetries from data or has them by construction. Two families dominate: atom-centred symmetry functions fed to atomwise neural networks [2], [3] and kernel methods on descriptors of atomic environments [4]; earlier work on molecules used the sorted eigenvalue spectrum of the Coulomb matrix to remove the atom-ordering dependence [5]. Learned invariant and equivariant message-passing networks have since become standard [6], [7].

We ask a deliberately small question on a system with an analytic reference: for Lennard-Jones (LJ) clusters of 5–13 atoms, how do (a) raw coordinates, (b) sorted pair distances and (c) sum-pooled symmetry functions compare when combined with kernel ridge regression (KRR) or a small multilayer perceptron (MLP), in terms of energy error, data efficiency and behaviour when the test set is rotated or its atoms are permuted? We pre-registered four hypotheses (Section IV). The study is a reimplementation-scale comparison on CPU; we claim no new method.

Contributions: (i) a controlled comparison with identical data, splits and tuning budget for all seven systems and a mean-predictor reference; (ii) learning curves from 50 to 1500 training configurations; (iii) measured invariance defects under rotation and permutation; (iv) three ablations (angular functions, linear ridge instead of the kernel, training-time augmentation) and two sensitivity checks (KRR hyperparameter grid, MLP training length). The outcome is mixed: our pre-registered method, symmetry-function KRR, is second to the sorted-distance MLP at $n=500$, raw coordinates give no useful model, and the ranking of symmetry-function KRR against sorted-distance KRR depends on the width of the hyperparameter grid (Section VI), which the first version of this study got wrong.

II. Related Work

Neural and kernel potentials. Behler and Parrinello introduced neural-network potentials in which the energy is a sum of atomic contributions computed from atom-centred symmetry functions of the local environment [2]; the radial and angular function families and their cutoff are detailed in [3], and we use that construction. The

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summed-atomic-energy form is an instance of the sum-decomposition that characterises permutation-invariant functions on sets [8]. Bartók et al. review representations of atomic neighbourhoods and propose the SOAP similarity for kernel regression [4]; we use a plain RBF kernel on symmetry-function vectors instead. Barthelmé et al. show that Gaussian-kernel regression tends to polynomial regression in the flat limit of very wide kernels, and that the best predictions may lie at length-scales that are numerically hard to reach [9]; our symmetry-function KRR turned out to sit in this regime.

Ordering-free global representations. Rupp et al. encode molecules with a Coulomb matrix and remove its dependence on atom ordering by using the sorted eigenvalue spectrum as the descriptor for kernel ridge regression [5]. Our sorted-distance descriptor follows the same idea of a canonical ordering, applied directly to the list of inverse pair distances of a single-element cluster.

Comparisons and reviews. Zuo et al. compare descriptors and regressors for interatomic potentials in terms of accuracy and cost [10], and Parsaeifard et al. assess the structural resolution of fingerprints used in machine learning of atomistic systems [11]; a broad review of machine-learned force fields is [1]. SchNet [6] and NequIP [7] learn invariant or equivariant features end to end and are out of scope here. The minima of LJ clusters are tabulated in [12]. Relative to these works we offer only a small reproducible comparison, not a new representation.

III. Method

Reference energy. Reduced units $\epsilon=\sigma=1$: $E(\mathbf{x}) = \sum_{i<j} 4(r_{ij}^{-12} - r_{ij}^{-6})$, $r_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$. Clusters have $N \in \{5, \dots, 13\}$ atoms, zero-padded to 13.

Descriptors. (a) Raw: $\phi_{\text{raw}} = [\text{vec}(\mathbf{X}), N]$ with atoms in random order. (b) Sorted: $\phi_{\text{sort}} = [\text{sort}_{\downarrow}(\{1/r_{ij}\}_{i<j}), N]$, zero-padded to 78 entries (zero stands for infinite distance); it is exactly invariant to rotation, translation and permutation. (c) Symmetry functions (SF): per atom i with cosine cutoff $f_c(r) = \frac{1}{2}[\cos(\pi r/R_c) + 1]$ for $r < R_c=4$ and 0 otherwise,

$$G_i^2 = \sum_{j \neq i} e^{-\eta(r_{ij}-r_s)^2} f_c(r_{ij}), \quad (1)$$

$$G_i^4 = 2^{1-\zeta} \sum_{\substack{j,k \neq i \\ j \neq k}} (1 + \lambda \cos \theta_{jik})^{\zeta} e^{-\eta(r_{ij}^2 + r_{ik}^2 + r_{jk}^2)} \times f_c(r_{ij}) f_c(r_{ik}) f_c(r_{jk}), \quad (2)$$

with $r_s \in \{0.9, 1.1, 1.4, 1.8, 2.3, 3.0\}$, $\eta \in \{2, 6\}$ (12 radial functions), a coordination number $\sum_{j \neq i} f_c(r_{ij})$, and eight angular functions ($\eta \in \{0.3, 1.5\}$, $\zeta \in \{1, 4\}$, $\lambda = \pm 1$): 21 functions per atom. Sum pooling gives the global descriptor $\phi_{\text{SF}} = [\sum_i \mathbf{g}_i, N]$, invariant to permutation by construction and to rotation because the $\mathbf{g}_i$ depend on distances and angles only. Features are standardised with training statistics.

KRR (reimplemented). With standardised descriptor $\mathbf{z} \in \mathbb{R}^d$, $k(\mathbf{z}, \mathbf{z}') = \exp(-\gamma \|\mathbf{z} - \mathbf{z}'\|^2/d)$ and

$$\hat{E}(\mathbf{z}) = \bar{E} + s \sum_m \alpha_m k(\mathbf{z}, \mathbf{z}_m), \quad \alpha = (K + \lambda I)^{-1} \tilde{\mathbf{y}}, \quad (3)$$

$\tilde{\mathbf{y}} = (\mathbf{y} - \bar{E})/s$ ($\bar{E}, s$: training mean and standard deviation). The system is solved by Cholesky factorisation; a (γ, λ) pair is skipped if the factorisation fails or LAPACK reports the matrix as ill-conditioned (reciprocal condition number below machine precision).

Linear ridge (ablation). The same estimator with the linear kernel $k(\mathbf{z}, \mathbf{z}') = \mathbf{z}^\top \mathbf{z}'/d$, i.e. ridge regression on the standardised descriptor; only λ is selected.

MLP (reimplemented). $\hat{E} = W_3 \text{SiLU}(W_2 \text{SiLU}(W_1 \phi + b_1) + b_2) + b_3$, widths 128, trained with Adam on the mean squared error of standardised energies.

Atomwise network (BP-style, reimplemented). $\hat{E} = \sum_{i=1}^N \text{NN}(\mathbf{g}_i)$ with the same 2×128 SiLU network shared over atoms (a single element), trained on the total-energy squared error; this is the Behler-Parrinello form [2].

Mean predictor. Reference that outputs the training mean energy. The method pre-registered as ours is SymFn-sum KRR ($\phi_{\text{SF}} + \text{KRR}$); the other six systems and the mean predictor are baselines or references.

IV. Experimental Setup

Data. A fixed library holds up to four of the lowest distinct local minima per size, found by L-BFGS from 40 random starts per N : one minimum for $N=5$, two for $N=6$ and four for each $N \geq 7$ (the lowest 13-atom energy found, -44.327 , agrees with the published global minimum [12]). Each configuration has a uniformly random N ; half are a library minimum with a random rotation plus Gaussian noise (std $\sim U[0.02, 0.12]$ per coordinate), half are random points in a ball of radius $0.62N^{1/3} + 0.35$ with pair distance ≥ 0.85 . We resample until $E \leq 50$. This truncation was chosen before the main runs because the perturbed-minimum energies are heavy-tailed: in a logged check of 4000 untruncated samples, a fraction 0.06600 of the perturbed configurations exceeds 50, with a maximum of 5355, against a fraction 0.001000 and a maximum of 65.94 for the random ones. Atom order is random. The minima library is shared by train and test, so test points are not from unseen minima.

Splits and seeds. For seed s all systems see the same n training, $\max(50, \lfloor n/4 \rfloor)$ validation (hyperparameter selection only) and 1000 test configurations. Main group: $n=500$, seeds 0–4. Learning curve: $n \in \{50, 150, 1500\}$, seeds 0–2 (the $n=500$ point uses the main runs). The three sets are drawn in sequence from one generator per seed, so for a given seed the training sets are nested across n but the validation and test sets differ between sizes. Ablations: $n=500$, seeds 0–4 for KRR and ridge variants, seeds 0–2 for MLP variants. Test-set transforms: random rotation, random permutation, and both (fixed transform seed per run).

Metrics. Test energy MAE (units of ϵ ; the standard deviation of the test energies is between 15.24 and 16.15 over all runs), MAE per atom, MAE on perturbed and random subsets, MAE on rotated+permuted test inputs, and the invariance defect $|\hat{E}(g\mathbf{x}) - \hat{E}(\mathbf{x})|$, the test-set mean for one random rotation+permutation g per configuration.

Tuning budget (validation only). For KRR we take the pair with the lowest validation MAE among 19 values of γ (10^{-7} to 10^2 in half decades) and 16 values of λ (10^{-13} to 10^2 in decades), identically for every descriptor. Linear ridge: the same λ values. MLP: 3000 Adam steps (batch 64, cosine schedule), learning rate in $\{10^{-3}, 4 \cdot 10^{-3}\}$. Architecture and steps are fixed, not tuned per system; the mean predictor is untuned. The KRR grid was widened after an external audit of the first version of this study, which had used a 5×5 grid ($\gamma \in [10^{-2}, 1]$, $\lambda \in [10^{-9}, 10^{-1}]$); the new bounds were set from validation errors only and all KRR runs were repeated (Section VI).

Hardware and runtime. Shared CPU machine, 2 threads, PyTorch CPU, two runs at a time. The slowest KRR or ridge run took 14.73 s, the slowest MLP run 176.0 s and the slowest atomwise run 182.0 s (in-run wall-clock time, which is not reproducible; the spread within a kind is mostly machine load). The MLP runs were not repeated, since their code path did not change.

Hypotheses (registered in proposal.md before the runs). H1: both invariant descriptors beat raw coordinates at $n=500$ for both regressors (Welch test, 5 seeds). H2: raw-coordinate models degrade at least $2\times$ under rotation/permutation, invariant ones are unchanged (defect $< 10^{-4}$). H3: SF-KRR beats sorted-distance KRR at $n \in \{50, 150\}$ (Welch test, 3 seeds). H4: KRR beats the MLP of the same descriptor at $n \leq 150$ for the majority of descriptors, and the gap shrinks or reverses at larger n . All tests are two-sided Welch t -tests at $\alpha = 0.05$ without multiple-comparison correction. Which statistics are printed. A p -value is printed only where rh compare recomputes it from the run registry, which it does within the main group ($n=500$) with SymFn-sum KRR as the reference (Table III). The registered tests at the other training sizes, between other pairs of systems, and between an ablation and its full model are computed by experiments/analyze.py and stored in results/analysis_stats.txt in the repository; for these we state the outcome but print no p -value, and give the registry means and the per-seed ranges (smallest to largest seed value) instead.

V. Results

Overview. Table I gives the results at $n=500$. Sorted-distance MLP is best (0.635), followed by our pre-registered SymFn-sum KRR (0.889), sorted-distance KRR (1.095), the atomwise BP-style network (3.130), SymFn-sum MLP (4.513), and the raw-coordinate models (12.178 KRR, 15.028 MLP) against 12.203 for the mean predictor.

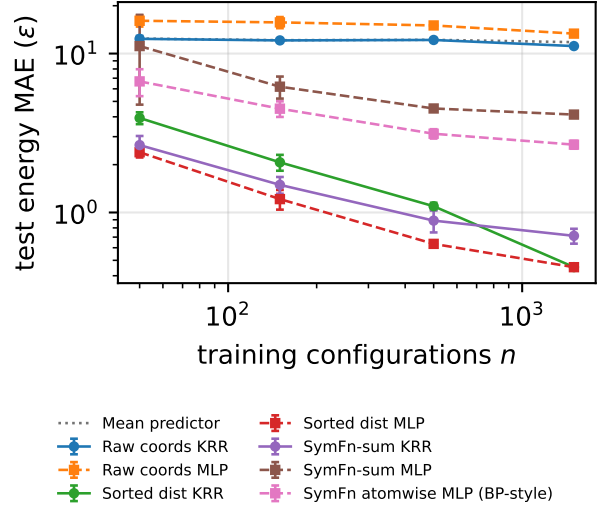


Fig. 1. Learning curves (test energy MAE versus training set size, mean $\pm$ std over 3 seeds; 5 seeds at $n=500$). Solid: KRR, dashed: neural networks, dotted: mean predictor.

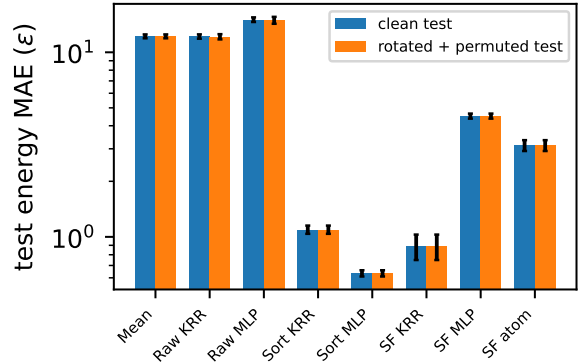


Fig. 2. Energy MAE on the clean test set and on the rotated+permuted test set, $n=500$ (log scale). Invariant descriptors give identical bars.

Over 5 seeds SymFn-sum KRR is worse than the sorted-distance MLP (Welch $p = 0.01427$, Table III) and better than sorted-distance KRR ($p = 0.02647$); its seed standard deviation (0.1399) exceeds that of both sorted-distance systems.

H1 (invariant descriptors beat raw): supported. The four registered comparisons at $n=500$ all pass the registered test, with large margins (the closest is SymFn-sum MLP 4.513 versus raw MLP 15.028). Table III prints the one that rh compare recomputes (SymFn-sum KRR versus raw KRR, $p = 3.210 \times 10^{-9}$); for the other three the per-seed ranges are far apart: sorted-distance KRR 1.040–1.188 against raw KRR 11.64–12.47, and sorted-distance MLP 0.6122–0.6736 and SymFn-sum MLP 4.330–4.679 against raw MLP 14.74–15.78. The same direction holds at the other sizes, where only SymFn-sum MLP versus raw

TABLE I

Main results, $n=500$ training configurations, mean $\pm$ std over 5 seeds (energy units ϵ ; lower is better). “perturbed”/“random”: MAE on the two test subsets; “rot+perm”: MAE after random rotation and atom permutation of the test inputs; “inv. defect”: test-set mean of $|\hat{E}(g\mathbf{x}) - \hat{E}(\mathbf{x})|$, averaged over seeds, and its largest per-seed value.

System	MAE	MAE/atom	perturbed	random	rot+perm	inv. defect	max over seeds
Mean predictor	12.20 $\pm$ 0.2953	1.307 $\pm$ 0.02987	15.02 $\pm$ 0.7511	9.387 $\pm$ 0.7936	12.20 $\pm$ 0.2953	0	0
Raw coords KRR	12.18 $\pm$ 0.3289	1.315 $\pm$ 0.03386	14.40 $\pm$ 0.6220	9.956 $\pm$ 1.061	12.14 $\pm$ 0.3958	2.881	6.967
Raw coords MLP	15.03 $\pm$ 0.4327	1.674 $\pm$ 0.05612	15.27 $\pm$ 0.4372	14.79 $\pm$ 0.9650	14.92 $\pm$ 0.6438	15.23	15.78
Sorted dist KRR	1.095 $\pm$ 0.05625	0.1208 $\pm$ 0.006047	1.050 $\pm$ 0.06309	1.140 $\pm$ 0.06707	1.095 $\pm$ 0.05625	6.230×10^{-11}	1.194×10^{-10}
Sorted dist MLP	0.6347 $\pm$ 0.02465	0.06970 $\pm$ 0.001850	0.6011 $\pm$ 0.05259	0.6682 $\pm$ 0.02875	0.6347 $\pm$ 0.02465	0	0
SymFn-sum KRR	0.8890 $\pm$ 0.1399	0.09288 $\pm$ 0.01314	0.9400 $\pm$ 0.1672	0.8379 $\pm$ 0.1370	0.8890 $\pm$ 0.1399	6.160×10^{-5}	8.340×10^{-5}
SymFn-sum MLP	4.513 $\pm$ 0.1422	0.5033 $\pm$ 0.01498	3.639 $\pm$ 0.1701	5.387 $\pm$ 0.2527	4.513 $\pm$ 0.1422	0	0
SymFn atomwise MLP (BP-style)	3.130 $\pm$ 0.2118	0.3435 $\pm$ 0.01968	2.605 $\pm$ 0.2966	3.655 $\pm$ 0.2665	3.130 $\pm$ 0.2118	5.174×10^{-7}	5.389×10^{-7}

TABLE II

Test energy MAE (ϵ) versus training size, mean $\pm$ std over seeds (3 seeds; 5 at $n=500$). SF: SymFn-sum.

System	$n=50$	$n=150$	$n=500$	$n=1500$
Mean pred.	12.43 $\pm$ 0.7970	12.14 $\pm$ 0.3877	12.20 $\pm$ 0.2953	11.80 $\pm$ 0.3456
Raw KRR	12.38 $\pm$ 0.5351	12.10 $\pm$ 0.2760	12.18 $\pm$ 0.3289	11.14 $\pm$ 0.2397
Raw MLP	16.08 $\pm$ 1.299	15.71 $\pm$ 1.265	15.03 $\pm$ 0.4327	13.35 $\pm$ 0.2601
Sorted KRR	3.933 $\pm$ 0.3371	2.068 $\pm$ 0.2369	1.095 $\pm$ 0.05625	0.4531 $\pm$ 0.02087
Sorted MLP	2.394 $\pm$ 0.1758	1.214 $\pm$ 0.1719	0.6347 $\pm$ 0.02465	0.4527 $\pm$ 0.009125
SF KRR	2.655 $\pm$ 0.3749	1.492 $\pm$ 0.1804	0.8890 $\pm$ 0.1399	0.7131 $\pm$ 0.07660
SF MLP	11.15 $\pm$ 6.376	6.188 $\pm$ 0.9792	4.513 $\pm$ 0.1422	4.131 $\pm$ 0.1574
SF atomwise MLP	6.683 $\pm$ 1.287	4.499 $\pm$ 0.5068	3.130 $\pm$ 0.2118	2.672 $\pm$ 0.1525

TABLE III

Two-sided Welch tests of SymFn-sum KRR (SF KRR, MAE 0.8890) against every other system B on test MAE, $n=500$, 5 seeds, as recomputed from the run registry by rh compare: mean MAE of B, difference of the means and p -value. Tests at other sizes and between other pairs are not printed (Section IV).

System B	MAE of B	SF KRR – B	Welch p
Mean predictor	12.20	−11.31	7.574×10^{-10}
Raw coords KRR	12.18	−11.29	3.210×10^{-9}
Raw coords MLP	15.03	−14.14	1.971×10^{-8}
Sorted dist KRR	1.095	−0.2060	0.02647
Sorted dist MLP	0.6347	0.2543	0.01427
SymFn-sum MLP	4.513	−3.624	1.491×10^{-10}
SymFn atomwise MLP	3.130	−2.241	2.379×10^{-7}

MLP at $n=50$ is unresolved (seed ranges 6.885–18.48 and 14.92–17.48). The comparison is partly trivial, however: raw KRR is not distinguishable from the mean predictor at any size by the Welch test (12.178 versus 12.203 at $n=500$, 11.14 versus 11.80 at $n=1500$), and the raw MLP is worse than the mean predictor at every size (Table II).

H2 (rotation/permutation behaviour): split verdict. Invariant descriptors are unchanged: the rot+perm column equals the clean MAE for all sorted-distance and SF systems, and the largest per-seed invariance defect of any invariant system is $8.34 \cdot 10^{-5}$ (SymFn-sum KRR), below the registered 10^{-4} but not by a wide margin. The defect is exactly 0 for the sum-pooled SF MLP and the sorted-distance MLP, $5.39 \cdot 10^{-7}$ at most for the atomwise MLP and $1.19 \cdot 10^{-10}$ for sorted-distance KRR. The SymFn-sum KRR value is a mean over test configurations; single configurations change by up to $4.2 \cdot 10^{-3}$ in the main runs so this model is invariant only up to an error that grows as the selected kernel becomes flatter (Section VI).

The “degrades at least $2\times$ ” prediction for raw models is refuted: raw KRR goes from 12.178 to 12.143 and raw MLP from 15.028 to 14.918 in Table I, which is expected for models that are no better than a constant. What the raw models do show is a large prediction inconsistency: their defect is 2.881 (KRR) and 15.23 (MLP).

H3 (SF-KRR beats sorted-distance KRR at small n): supported, with caveats. At $n=50$ SF-KRR has MAE 2.655 versus 3.933 for sorted KRR and at $n=150$ 1.492 versus 2.068 (Table II); both registered tests pass at $\alpha = 0.05$, and the per-seed ranges are disjoint at both sizes (2.247–2.984 against 3.610–4.283, and 1.364–1.699 against 1.795–2.214). Three caveats limit this. First, each test has 3 seeds and the two tests are uncorrected; the $n=150$ test would not pass a Bonferroni correction for the two tests. Second, the advantage does not persist: it is still present at $n=500$ ($p = 0.02647$, Table III) but reversed at $n=1500$, where sorted KRR reaches 0.4531 and SF-KRR 0.7131 (disjoint seed ranges 0.4364–0.4765 and 0.6336–0.7864). Third, the verdict depends on the hyperparameter grid: with the narrow grid of our first version SF-KRR is worse than sorted KRR at $n=500$ (1.587 versus 1.096, Table IV), and that version reported H3 as not supported. H3 concerns KRR only; the sorted-distance MLP (2.394 and 1.214 at $n=50$ and 150) has a lower mean error than SF-KRR at both sizes (not resolved; the seed ranges overlap), so the data do not show that symmetry functions are the more data-efficient descriptor in general.

H4 (KRR beats MLP at small n): mixed, descriptor dependent. For raw input KRR is lower at every size (Table II), but KRR is at the level of the mean predictor and the MLP is above it, so this is not informative. For SF-sum input KRR is lower at every size, and the difference is resolved for $n \geq 150$ ($p = 1.491 \times 10^{-10}$ at $n=500$, Table III); at $n=50$ the MLP has one poor seed (std 6.376) and the difference is not resolved. For sorted distances the MLP is better at $n=50, 150, 500$ and the two tie at $n=1500$ (0.4531 versus 0.4527): the gap shrinks with data as H4 states, but its sign at small n is opposite for this descriptor. By the registered majority criterion (two of three descriptors have KRR ahead at $n \leq 150$) H4 is nominally supported, which we regard as weak, since one of the two is the uninformative raw case, and for SF input

TABLE IV

KRR hyperparameter-grid sensitivity, $n=500$, 5 seeds. Narrow: $\gamma \in [10^{-2}, 1]$, $\lambda \in [10^{-9}, 10^{-1}]$ (5×5 values, first version); mid: $\gamma \in [10^{-4}, 1]$, $\lambda \in [10^{-11}, 10^{-1}]$; wide: the grid of all other tables. Last rows: number of seeds whose selected γ is the first or last grid value, and whose selected λ is the smallest feasible one.

System	narrow	mid	wide (main)
SF KRR	1.587 ± 0.122	0.907 ± 0.099	0.889 ± 0.140
inv. defect	$1.13 \cdot 10^{-6}$	$1.58 \cdot 10^{-5}$	$6.16 \cdot 10^{-5}$
γ at edge / λ at floor	4/5, 5/5	0/5, 5/5	0/5, 5/5
Sorted KRR	1.096 ± 0.058	1.095 ± 0.056	1.095 ± 0.056
inv. defect	$8.05 \cdot 10^{-11}$	$6.10 \cdot 10^{-11}$	$6.23 \cdot 10^{-11}$
γ at edge / λ at floor	2/5, 3/5	2/5, 3/5	0/5, 3/5

TABLE V

Selected KRR hyperparameters with the wide grid (range over seeds; raw KRR pooled over the four sizes) and number of runs with γ on a grid edge or λ on the smallest feasible value for the selected γ .

System	n	γ	λ	γ edge	λ floor
Raw KRR	all	$10^{-0.5-10^1}$	10^{-13-10^1}	0/14	3/14
Sorted KRR	50	$10^{-0.5-10^0}$	$10^{-13-10^{-4}}$	0/3	1/3
Sorted KRR	150	$10^{-1-10^{-0.5}}$	$10^{-5-10^{-4}}$	0/3	0/3
Sorted KRR	500	$10^{-0.5-10^0}$	$10^{-13-10^{-4}}$	0/5	3/5
Sorted KRR	1500	$10^{-0.5}$	$10^{-9-10^{-6}}$	0/3	0/3
SF KRR	50	$10^{-7-10^{-6}}$	$10^{-13-10^{-12}}$	1/3	2/3
SF KRR	150	$10^{-3.5}$	10^{-12}	0/3	3/3
SF KRR	500	$10^{-4-10^{-3.5}}$	10^{-12}	0/5	5/5
SF KRR	1500	$10^{-3.5-10^{-3}}$	10^{-11}	0/3	3/3
SF KRR radial-only	500	$10^{-3.5-10^{-2.5}}$	$10^{-11-10^{-10}}$	0/5	0/5
SF linear ridge	500	—	$10^{-10-10^{-9}}$	—	0/5
SF lin. ridge rad.-only	500	—	$10^{-10-10^{-9}}$	—	0/5
Sorted linear ridge	500	—	$10^{-3-10^{-1}}$	—	0/5

the MLP error remains several times the KRR error at every size (Table II).

Where it fails. (i) The two networks on symmetry functions are the clearest failure: SymFn-sum MLP has 4.513 at $n=500$ and 4.13 at $n=1500$, and the atomwise network 3.130 and 2.67, while plain linear ridge regression on the same pooled descriptor reaches 0.805 at $n=500$ (Table VI). The descriptor therefore carries the information and our fixed network training does not extract it. Both networks are worse on random than on perturbed-minimum configurations (5.387 versus 3.639 and 3.655 versus 2.605 in Table I), unlike the sorted-distance MLP (0.668 versus 0.601). (ii) SF-KRR saturates: it improves from 0.8890 to 0.7131 between $n=500$ and 1500 while sorted KRR improves from 1.095 to 0.4531, so with the most data it is the least accurate of the three accurate systems. A limited resolution of the 21 pooled functions may be the cause; we did not vary the number of functions. (iii) We did not examine per-size errors, so we cannot say whether any N dominates.

VI. Ablations and Analysis

KRR hyperparameter grid. Table IV repeats the two invariant KRR systems with three grids. Sorted-distance KRR is insensitive (1.096, 1.095, 1.095). SymFn-sum KRR is not: 1.587 with the narrow grid, where 4 of 5 seeds select the smallest γ , against 0.907 and 0.889 with the mid and

TABLE VI

Ablations at $n=500$, test energy MAE, mean $\pm$ std over the listed number of seeds. The full models are the main-group runs (seeds 0–4); the MLP variants were run on seeds 0–2.

Variant	seeds	MAE	inv. defect
SymFn-sum KRR (full)	5	0.8890 ± 0.1399	6.160×10^{-5}
radial-only	5	0.7850 ± 0.07148	5.115×10^{-6}
linear ridge (no kernel)	5	0.8047 ± 0.06394	4.895×10^{-5}
linear ridge, radial-only	5	0.9138 ± 0.1093	2.874×10^{-5}
Sorted dist KRR (full)	5	1.095 ± 0.05625	6.230×10^{-11}
linear ridge (no kernel)	5	5.647 ± 0.2097	8.256×10^{-12}
SymFn-sum MLP (full)	5	4.513 ± 0.1422	0
radial-only	3	4.582 ± 0.1915	0
Raw coords MLP	5	15.03 ± 0.4327	15.23
+ rot/perm aug.	3	9.773 ± 0.4365	1.700

TABLE VII

Sensitivity of the MLPs to the number of Adam steps, $n=500$, test energy MAE, mean $\pm$ std. The 1000- and 12000-step runs use seeds 0–2; the 3000-step column is the main group (seeds 0–4).

System	1000 steps	3000 steps (main)	12000 steps
Sorted dist MLP	0.8914 ± 0.04570	0.6347 ± 0.02465	0.5388 ± 0.02375
seeds	3	5	3
SymFn-sum MLP	4.809 ± 0.2263	4.513 ± 0.1422	4.392 ± 0.1111
seeds	3	5	3

wide grids (per-seed ranges: narrow 1.440–1.750, disjoint from wide 0.7523–1.119; mid 0.7900–1.047, overlapping with wide). The narrow grid was therefore unfair to this one system; the wide grid is used everywhere else. With the wide grid the selected γ is interior in all but one run (Table V; SF-KRR, $n=50$, one seed at 10^{-7}). The selected λ is a different matter: SymFn-sum KRR takes the smallest value whose solve is not flagged as ill-conditioned in 13 of its 14 runs, and sorted-distance KRR in 4 of 14. We could not bracket it: smaller values give solves that LAPACK flags as ill-conditioned. The energies are noise-free, so the validation error may keep falling as $\lambda \rightarrow 0$ until round-off stops it. Two observations bound the consequence at $n=500$: widening from mid to wide changes the error only from 0.907 to 0.889, and linear ridge, whose λ is interior in all 5 seeds, gives a similar error (next paragraph). The price of the flat kernels is numerical: the invariance defect of SymFn-sum KRR rises from $1.13 \cdot 10^{-6}$ to $1.58 \cdot 10^{-5}$ to $6.16 \cdot 10^{-5}$ across the three grids, which may be round-off in the descriptor amplified by large dual coefficients; we did not measure the coefficients.

Linear ridge instead of the kernel. For SymFn-sum input, ridge regression without a kernel gives 0.805, not distinguishable from RBF KRR (0.889; the seed ranges 0.7177–0.8824 and 0.7523–1.119 overlap; Table VI); for sorted distances it gives 5.647 against 1.095. Together with a selected γ of at most 0.001000 in every SF-KRR run, this suggests that the selected SF kernels act almost linearly on the pooled descriptor, the flat-limit behaviour analysed in [9]. This is plausible for a pair potential, since E and the pooled radial functions are both sums over pairs of a

function of r_{ij} , but we did not test other potentials.

Angular symmetry functions. Dropping the eight angular functions gives 0.785 for KRR against 0.889 with them (overlapping seed ranges, 0.6936–0.8934 and 0.7523–1.119), 0.914 for linear ridge, and leaves the MLP unchanged within noise (4.582 on seeds 0–2 versus 4.513 for the full model on seeds 0–4). No effect of the angular terms is resolved in either direction. The first version of this study, with the narrow grid, reported a gain from removing them for KRR, which the wide grid does not reproduce.

Augmentation for raw coordinates. Training the raw MLP on freshly rotated and permuted minibatches lowers the MAE from 15.03 (seeds 0–4) to 9.773 (seeds 0–2; disjoint seed ranges 14.74–15.78 and 9.298–10.16), still far above any invariant descriptor, and the invariance defect drops from 15.23 to 1.700 but is not removed (Table VI). At this budget, augmentation is much less effective than building the symmetry into the descriptor.

MLP training length. Because the MLPs may be under-trained, we varied the step budget (Table VII). Sorted-distance MLP improves from 0.8914 to 0.6347 to 0.5388 over 1000, 3000 and 12000 steps (pairwise disjoint seed ranges 0.8444–0.9357, 0.6122–0.6736 and 0.5249–0.5662), so it was not converged at the budget of the main runs. SymFn-sum MLP goes from 4.809 to 4.513 to 4.392, differences that are not resolved (overlapping seed ranges 4.552–4.979, 4.330–4.679 and 4.314–4.519). Even with 12000 steps the SF-sum MLP (4.392) stays far from SF-KRR and linear ridge, so its gap is not closed by the step count within this range.

VII. Limitations

This is a small CPU study of one analytic system. (i) Single element, pair-only potential, at most 13 atoms; the near-linear behaviour of symmetry-function KRR and the descriptor ranking may not transfer to many-body or multi-element systems. (ii) The minima library (one to four minima per size) is shared by train and test, so we measure interpolation around known minima, not generalisation to new structures or sizes. (iii) Energies are truncated at $E \leq 50$, which excludes the strongly repulsive region. (iv) 3–5 seeds; tests with 3 seeds have little power, are uncorrected for multiplicity, and the H3 verdict rests on two such 3-seed tests. (v) Hyperparameters: the KRR grid was widened after an audit of the first version had shown the test-set consequences of the narrow grid, so the grid is not a pre-registered choice, although its bounds were set on validation error; the selected λ of symmetry-function KRR sits on the smallest numerically feasible value in 13 of 14 runs, and such nearly singular solves may be sensitive to the linear-algebra library. (vi) All baselines are our reimplementations; the MLP architecture and step count were not tuned per system, the step-count check shows the sorted-distance MLP was not converged, and the poor symmetry-function networks say more about

our training setup than about Behler–Parrinello networks. (vii) Raw-coordinate models were trained on randomly rotated, randomly ordered data, so they never had a chance to fit; training on canonically aligned data was not tried, and the H2 refutation applies to this setting only. (viii) Test sets differ between training sizes, so learning-curve slopes include test-sampling noise. (ix) Energies only: no forces, no dynamics, no per- N breakdown. (x) Invariance is measured with one random transform per test configuration; the exact zeros of two MLP systems may reflect their single-precision inputs. A fuller study would use unseen minima and sizes, forces, more seeds, tuned networks, SOAP-type kernels and equivariant networks.

VIII. Conclusion

On small LJ clusters, an invariant descriptor is necessary for a useful model in our setup: raw coordinates give errors at or above the mean-predictor level, and rotation or permutation then changes their predictions by 2.881 to 15.23 energy units on average. Among invariant systems at $n=500$, the sorted-distance MLP was the most accurate (0.635), symmetry-function KRR second (0.889) and sorted-distance KRR third (1.095). Symmetry-function KRR was ahead of sorted-distance KRR for $n \leq 500$ and behind it at $n=1500$, which supports the pre-registered small-data hypothesis for KRR on few seeds; the KRR-versus-MLP ranking depends on the descriptor. The symmetry-function result needed a much wider kernel-width range than the sorted-distance one, the selected kernels were nearly flat, linear ridge on the same descriptor did as well (0.805), and angular functions had no resolved effect (0.785 without them versus 0.889 with them). A shared hyperparameter grid that was adequate for one descriptor reversed the ranking of another. These findings come from one toy system, few seeds and reimplemented baselines, and are not evidence about production potentials.

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